



**CHRIST**  
(DEEMED TO BE UNIVERSITY)  
BANGALORE | DELHI NCR | PUNE

# Project Work **RM881**: 2026-27

## Vision-Assisted Dual-Arm Cable Handover and Insertion using MoveIt with Marker-Based Pose Estimation

### Mission

Christ University is a nurturing ground for an individual's holistic development to make effective contribution to the society in a dynamic environment

### Vision

Excellence and Service

### Core Values

Faith in God | Moral Uprightness  
Love of Fellow Beings | Social Responsibility  
Pursuit of Excellence

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# Introduction:

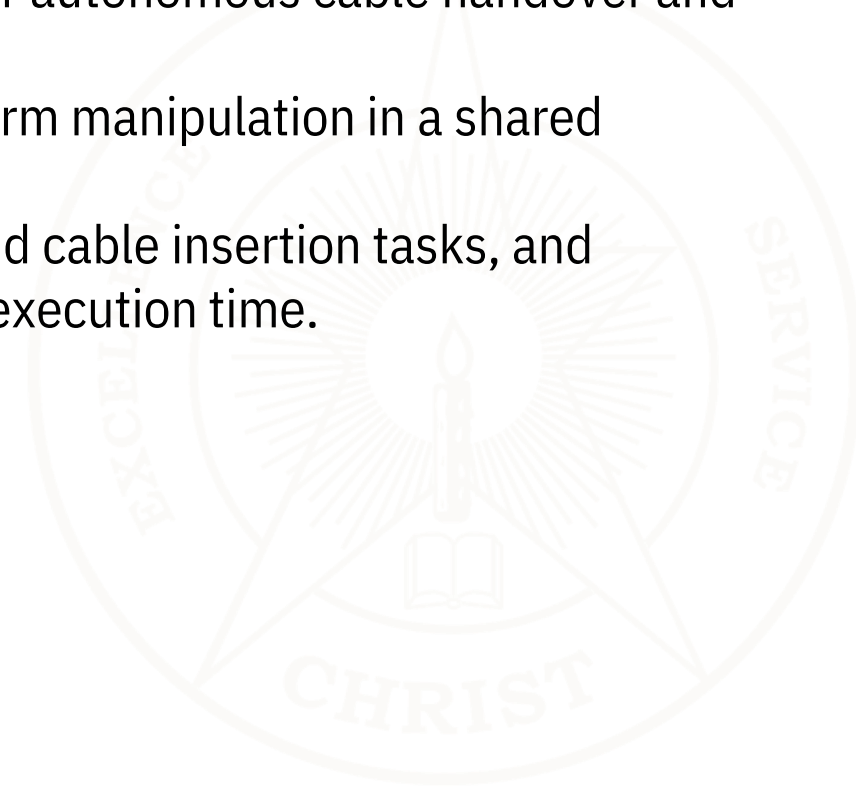
- Inserting flexible cables and connectors into housings is a common but largely manual task in electronics, automotive wiring-harness, and appliance assembly.
- It is naturally a two-arm task: one arm presents and stabilises the flexible cable while the other reaches the connector and performs the insertion, with a handover to re-grasp the cable nearer its tip.
- This project develops an autonomous dual-arm system that detects the cable grasp point and the target connector using fiducial (ArUco) markers, plans a collision-free handover in MoveIt, and performs a compliant insertion.
- The platform is a low-cost 7-DoF bio-inspired arm driven by ST3215 serial-bus servos, validated in simulation and on real hardware.

# Problem Identification/statement :

- Cable insertion is challenging because the cable is flexible (its shape and tip orientation change as it is handled), the connector must be reached at a precise pose, and the final insertion needs fine alignment.
- A single arm cannot both stabilise the cable mid-length and insert its tip; the task therefore requires coordinated dual-arm manipulation with an arm-to-arm handover.
- The two arms share one workspace, so trajectories must avoid inter-arm, arm-cable, and arm-environment collisions throughout pick, handover, and insertion.
- Problem: design a vision-assisted dual-arm system that detects cable and connector poses, plans a collision-free handover, and performs robust insertion tolerant of small pose and servo errors, validated on real low-cost hardware.

# Objectives:

- To develop an ArUco marker-based vision system for estimating the 3D pose of the cable, connector, and peg-in-hole components for robotic manipulation.
- To implement a MoveIt 2-based dual-arm coordination framework for autonomous cable handover and manipulation.
- To plan and execute collision-free coordinated trajectories for dual-arm manipulation in a shared workspace.
- To validate the dual-arm system through autonomous peg-in-hole and cable insertion tasks, and evaluate its success rate, handover reliability, positioning accuracy, and execution time.

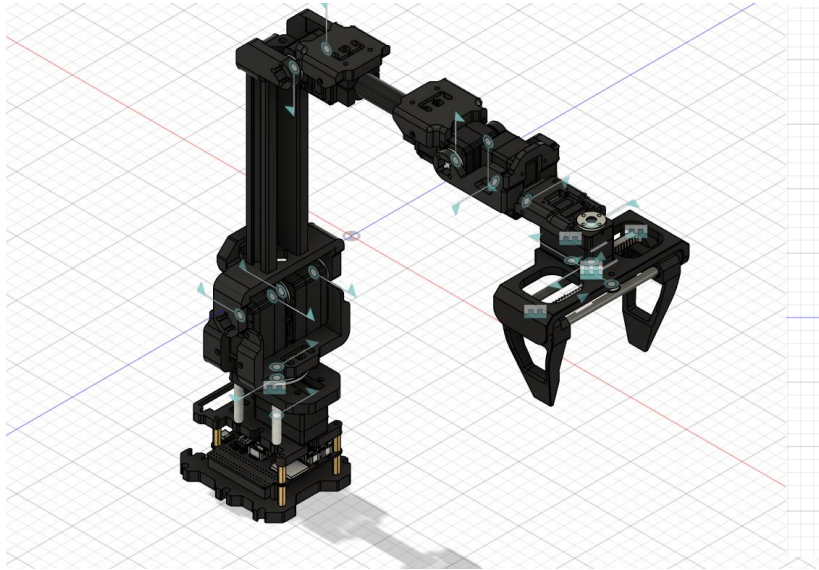


# Prototype Design (CAD):



- This was our first design iteration. Although it successfully demonstrated the basic concept, it revealed several mechanical and integration challenges during evaluation. Based on these findings, we redesigned the manipulator to improve strength, workspace, actuator integration, and overall performance, leading to our final prototype shown in the next slide.

# Prototype Design (CAD):



**Figure1:** CAD model of the arm.

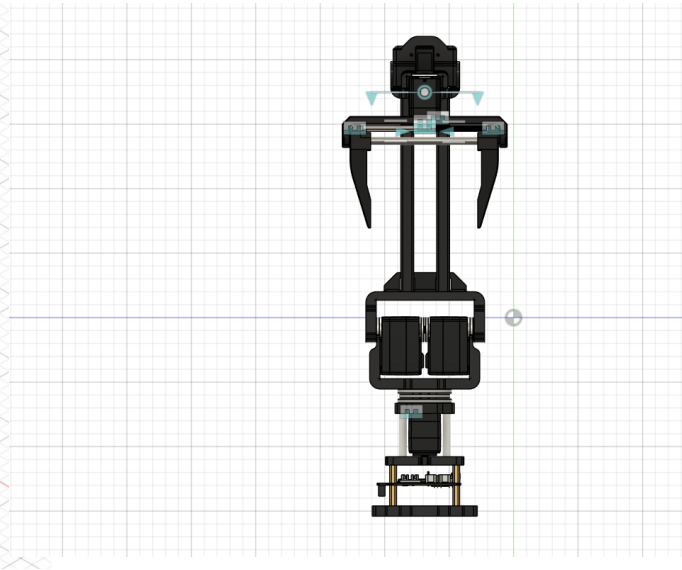


Figure2:Front view of the arm

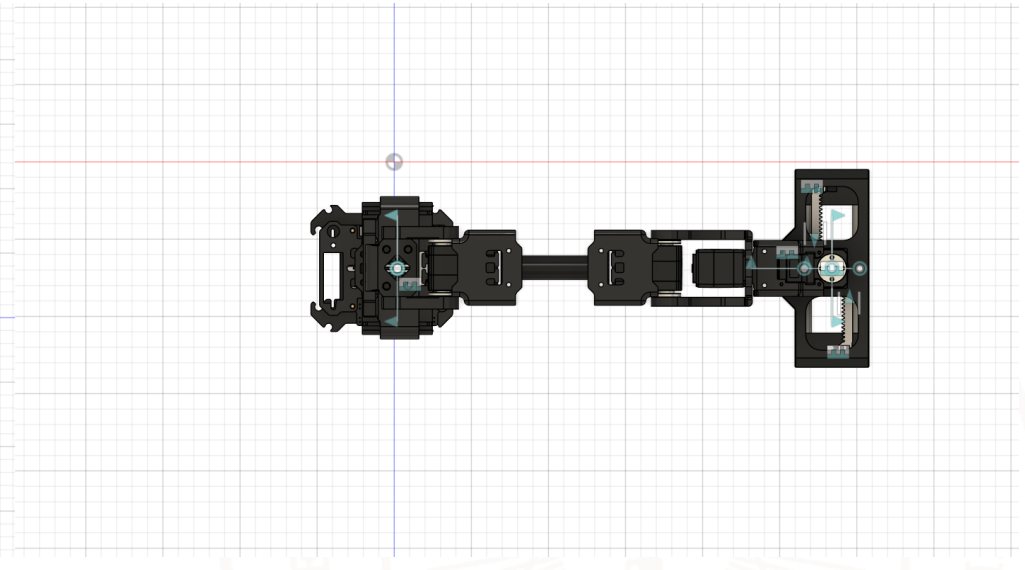


Figure3: Top view of the arm

- This redesigned prototype incorporates the improvements identified from our first design iteration. It offers better structural rigidity, a larger operating workspace, improved precision, and compatibility with ROS 2 and MoveIt 2, making it suitable for autonomous dual-arm cable insertion and peg-in-hole tasks.



# Upcoming work:

## MoveIt 2 Integration

- Configure the robot using the Move It Setup Assistant.
- Import the robot URDF/Xacro model into Move It 2.
- Define planning groups for the left arm, right arm, and dual-arm system.
- Configure end effectors (grippers), virtual joints, and collision matrices.
- Generate the Move It configuration package (SRDF, kinematics, controllers, and launch files).
- Integrate Move It with RViz for visualization and Gazebo for simulation.

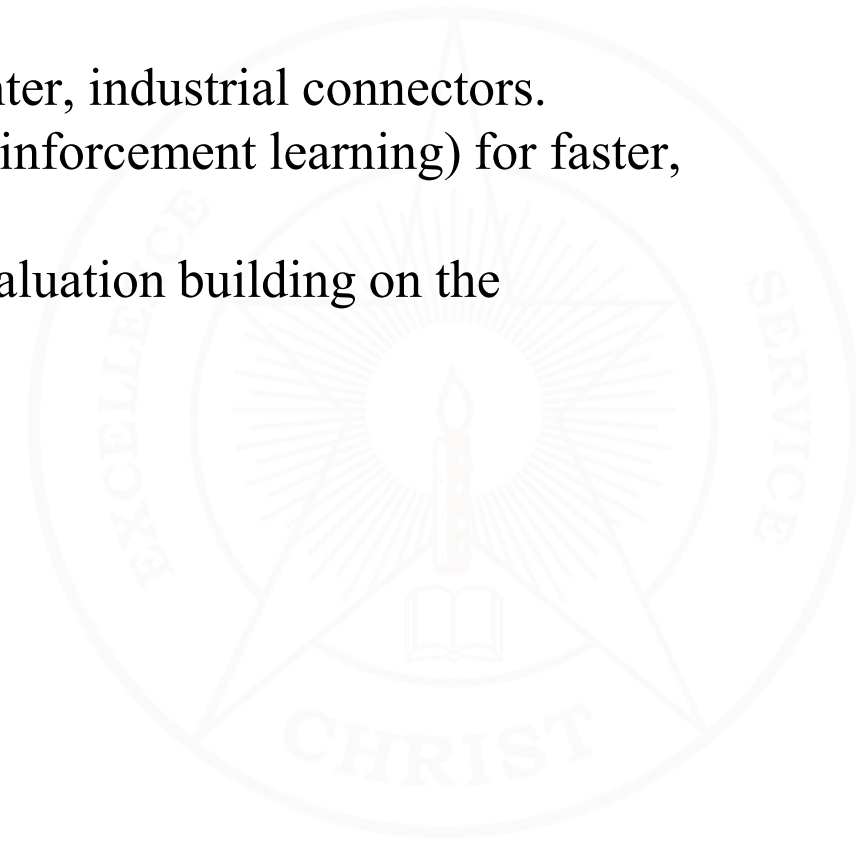
## Motion Planning

- Move It computes a **collision-free path** between the robot's current position and the target pose.
- The target pose is provided manually through RViz or automatically from the vision system.
- Inverse Kinematics (IK) calculates the required joint angles.
- The motion planner (OMPL) generates an optimal joint trajectory.
- Collision checking ensures the robot avoids self-collision, the support beam, the worktable, and surrounding obstacles.
- The planned trajectory is visualized in RViz before execution.



# Future Scope:

- Replace markers with a learned cable / connector detector (e.g. YOLO, keypoint network) for marker-free insertion.
- Add occlusion-robust cable shape estimation and tracking so the full deformable cable is followed during manipulation.
- Integrate force / tactile sensing for compliant, slip-aware insertion into tighter, industrial connectors.
- Learn the handover and insertion policy from demonstration (imitation / reinforcement learning) for faster, smoother transfers.
- Extend toward multi-cable wiring-harness assembly and a journal-level evaluation building on the conference results.



# Project Timeline

| Task                            | Jun 2026 | Jul 2026 | Aug 2026    | Sep 2026       | Oct 2026       | Nov 2026       |
|---------------------------------|----------|----------|-------------|----------------|----------------|----------------|
| Problem Statement and Objective | DONE     |          |             |                |                |                |
| CAD & URDF/XACRO                |          | DONE     | IN PROGRESS |                |                |                |
| ROS 2 & MoveIt 2                |          |          | IN PROGRESS | YET TO BE DONE |                |                |
| Vision Integration              |          |          |             | YET TO BE DONE | YET TO BE DONE |                |
| Testing & Evaluation            |          |          |             |                |                | YET TO BE DONE |
| Paper Drafting                  |          |          |             |                |                | YET TO BE DONE |



# Thank you